

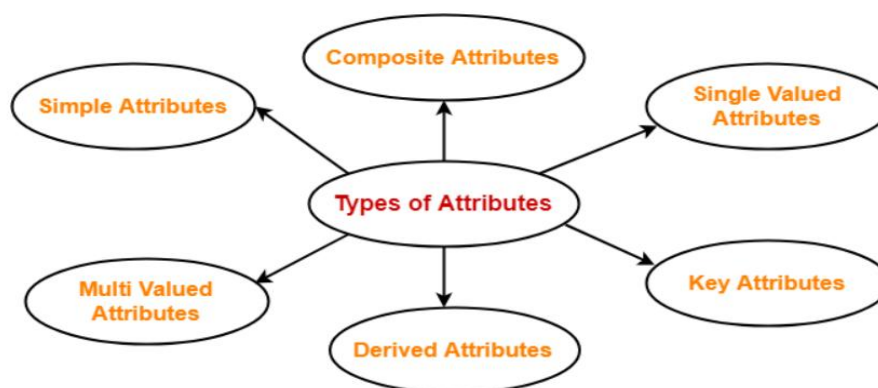
DBMS 4th Sem (PYQ + MP + Imp topics)

Module 1

- 1.Attributes and its types
- 2.Logical vs physical data independence.
- 3.Three schema architecture
- 4.Database and DBMS advantages.
- 5.Categories of data models.
- 6.Component modules of DBMS with diagram.
- 7.ER diagram

1. Attributes and its types

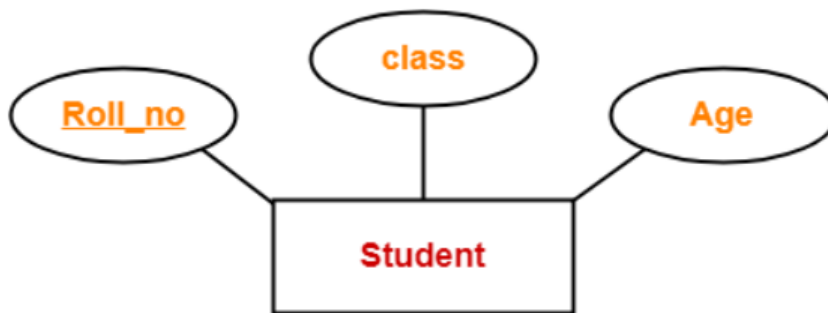
- An entity is represented by a set of attributes, that is descriptive properties possessed by all members of an entity set.



A. Simple Attribute

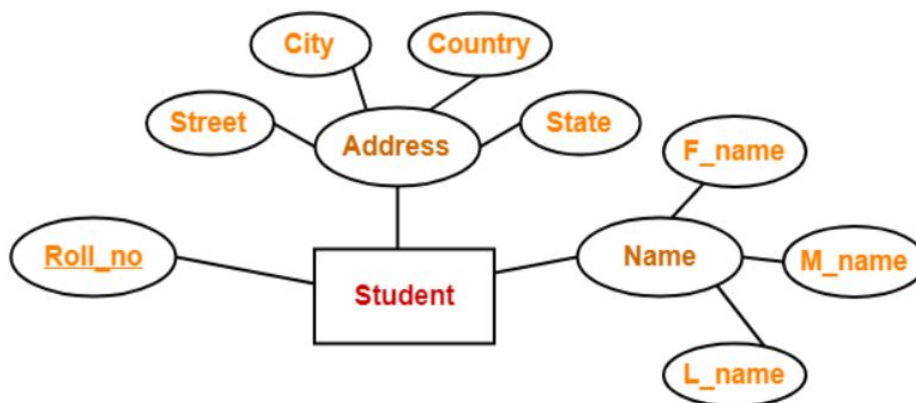
Simple attributes are those that are atomic (Not Divisible).

Example: Roll_no, Class and Age.



B. Composite Attributes

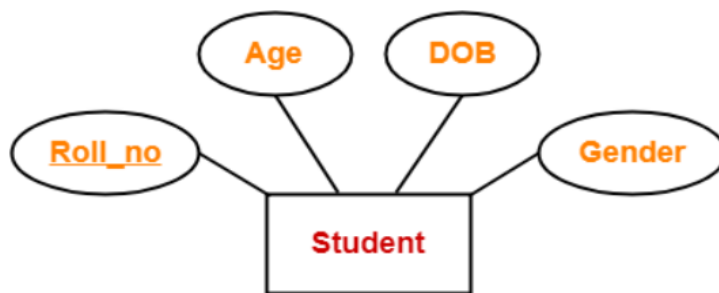
Composite attributes are those that can be further subdivided into simpler parts. Here, the attributes "Name" and "Address" are composite attributes as they are composed of many other simple attributes.



C. Single Valued Attribute

Single valued attributes are those attributes which can take only one value for a given entity from an entity set.

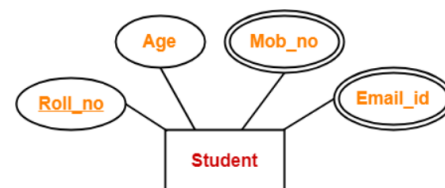
Here, all the attributes are single valued attributes as they can take only one specific value for each entity.



D. Multi-Valued Attributes

Multi valued attributes are those attributes which can take more than one value for a given entity from an entity set.

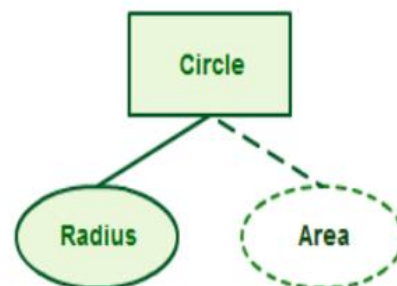
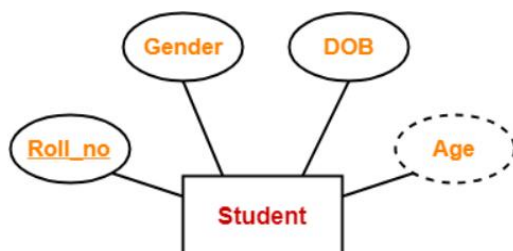
Here, the attributes "Mob_no" and "Email_id" are multi valued attributes as they can take more than one values for a given entity.



E. Derived Attributes

Derived attributes are those attributes which can be derived from other attribute(s).

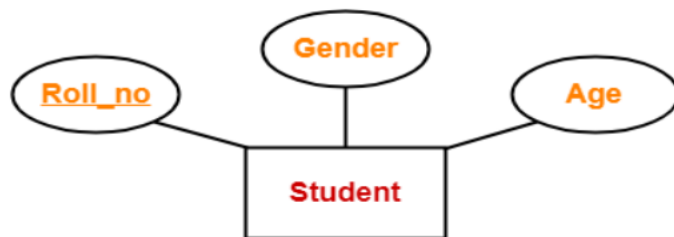
Here, the attribute "Age" is a derived attribute as it can be derived from the attribute "DOB".



F.Key Attributes

Key attributes are those attributes which can identify an entity uniquely in an entity set.

Here, the attribute “Roll_no” is a key attribute as it can identify any student uniquely.



2. Logical vs Physical data Independence

Logica Data Independence	Physical Data Independence
Logical Data Independence is mainly concerned with the structure or changing the data definition.	Mainly concerned with the storage of the data.
It is difficult as the retrieving of data is mainly dependent on the logical structure of data.	It is easy to retrieve.
Compared to Logic Physical independence it is difficult to achieve logical data independence.	Compared to Logical Independence it is easy to achieve physical data independence.
You need to make changes in the Application program if new fields are added or deleted from the database.	A change in the physical level usually does not need change at the Application program level.
Modification at the logical levels is significant whenever the logical structures of the database are changed.	Modifications made at the internal levels may or may not be needed to improve the performance of the structure.
Concerned with conceptual schema	Concerned with internal schema
Example: Add/Modify/Delete a new attribute	Example: change in compression techniques, hashing algorithms, storage devices, etc

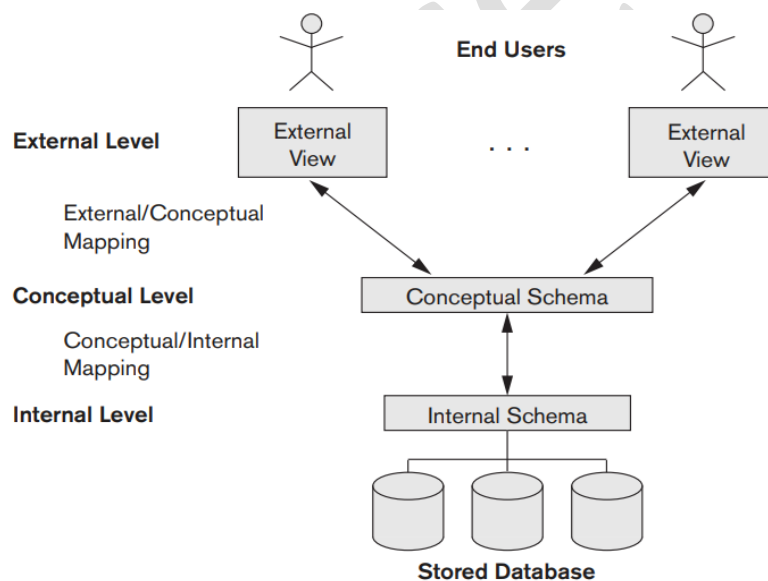
3. Three schema Architecture

Proposed to support DBMS characteristics of:

- Program-data independence.
- Support of multiple views of the data.

Not explicitly used in commercial DBMS products, but has been useful in explaining database system organization

- Defines DBMS schemas at *three* levels:
 - a. Internal schema at the internal level to describe physical storage structures and access paths (e.g indexes).
 - Typically uses a physical data model.
 - b. Conceptual schema at the conceptual level to describe the structure and constraints for the whole database for a community of users.
 - Uses a conceptual or an implementation data model.
 - c. External schemas at the external level to describe the various user views.
 - Usually uses the same data model as the conceptual schema.



4. Advantages Of DBMS Approach

1. • **Data Redundancy Control**
DBMS reduces duplication of data by storing it in a centralized database.
2. • **Data Consistency**
Since redundancy is controlled, the same data remains consistent across the system.

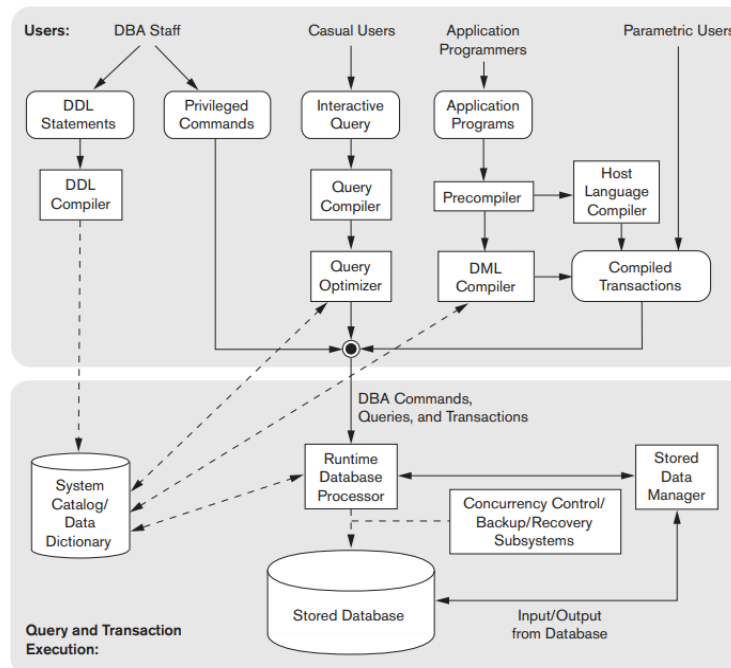
3. • **Data Sharing**
Multiple users and applications can access and share the same database simultaneously.
4. • **Data Security**
DBMS provides authorization and access control to protect sensitive data.
5. • **Data Integrity**
It ensures accuracy and reliability of data through constraints and rules.
6. • **Backup and Recovery**
DBMS provides mechanisms to recover data in case of system failure or data loss.

5. Categories of Data Modules

A set of concepts to describe the structure of a database, the operations for manipulating these structures, and certain constraints that the database should obey.

- Categories
 - Conceptual (high-level, semantic) data models:
 - Provide concepts that are close to the way many users perceive data.
 - (Also called *entity-based* or *object-based* data models.)
 - Physical (low-level, internal) data models:
 - Provide concepts that describe details of how data is stored in the computer. These are usually specified in an ad-hoc manner through DBMS design and administration manuals
 - Implementation (representational) data models:
 - Provide concepts that fall between the above two, used by many commercial DBMS implementations (e.g. relational data models used in many commercial systems).
 - Self-Describing Data Models:
 - Combine the description of data with the data values. Examples include XML, key-value stores and some NOSQL systems.

6. Component Modules Of DBMS



- The top part of the figure refers to the various users of the database environment and their interfaces.
- The lower part shows the internal modules of the DBMS responsible for storage of data and processing of transactions.
- The database and the DBMS catalog are usually stored on disk.
- Access to the disk is controlled primarily by the operating system (OS), which schedules disk read/write.
- Buffer management module - schedule disk read/write.
- A higher-level stored data manager module of the DBMS controls access to DBMS information that is stored on disk, whether it is part of the database or the catalog.
- The DBA staff works on defining the database and tuning it by making changes to its definition using the DDL and other privileged commands.
- Casual users and persons with occasional need for information from the database interact using the interactive query interface.

- Application programmers who create programs using some host programming languages
- Parametric users who do data entry work by supplying parameters to predefined transactions.
- The DDL compiler processes schema definitions, specified in the DDL, and stores descriptions of the schemas (meta-data) in the DBMS catalog.
- The catalog includes information such as the names and sizes of files, names and data types of data items, storage details of each file, mapping information among schemas, and constraints.
- Query compiler – compiles queries into an internal form.
- Query optimizer - concerned with the rearrangement and possible reordering of operations, elimination of redundancies, and use of efficient search algorithms during execution.
- Precompiler - extracts DML commands from an application program written in a host programming language
- Runtime database processor executes
 - (1) the privileged commands
 - (2) the executable query plans
 - (3) the canned transactions with runtime parameters.
- Concurrency control and backup and recovery systems are integrated into the working of the runtime database processor for purposes of transaction management.

The *relationship* can be described by certain characteristics.

- Degree of the relationship.
- Role.
- Recursive Relationship.
- Cardinality.
- Participation Constraint.

- Degree of The Relationship

The total number of entities participating in the relation is called degree of the Relationship.

1.Unary Relationship.

2. Binary Relationship.

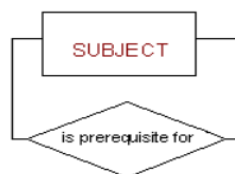
3. Ternary Relationship.

- Unary Relationship

When there is a relationship between two entities of the same type, it is known as a unary or recursive relationship. This means that the relationship is between different instances of the same entity type.

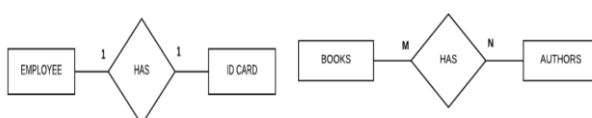
When both participants in the relationship are the same entity.

Ex: Subjects may be prerequisites for other subjects



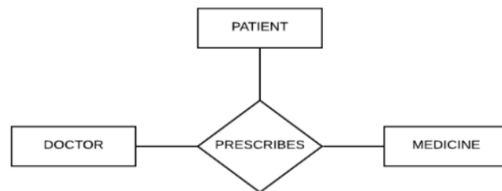
- Binary Relationship

When there is a relationship between two different entities, it is known as a binary relationship.



- Ternary Relationship

When there is a relationship between three different entities, it is known as a ternary relationship. An example of a ternary relationship can be shown as follows



- **Role**

- Role is the function an entity plays in the relationship.
- Roles are optional.
- They help in clarifying the semantics of a relationship.
- **EMPLOYEE** entity is playing the role of worker.



- **Recursive Relationship**

- When the relationship occurs within the same entity, it is called Recursive Relationship.
- An entity related to itself is the recursive relationship.
- The same entity performs different roles.
- **EMPLOYEE** entity is playing the role of worker.
- **EMPLOYEE** entity is playing the role of manager.

